

Velocity Fields of Optimal Transport Reveals Causality: Root Cause Analysis via Flow Matching

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Abstract

The problem of root cause analysis (RCA) is to identify the source of an anomaly in a system of multiple variables from observational data. The observed joint distribution is compatible with multiple causal structures, making RCA ill-posed without structural assumptions on the data generating process. In this work, we present a new source of asymmetry between cause and effect variables that arises when a system transitions from a normal to a faulty state. We show that this asymmetry is encoded in the velocity field of the optimal transport between the two states: the velocity of the root cause is unconstrained, while the velocity of downstream variables is mediated by the causal mechanism. We prove that this velocity field factorizes in the causal direction and cannot factorize in the reverse for any nonlinear mechanism, yielding a new identifiability result without requiring a known causal graph, bidirectional model fitting, or assumptions on the noise distribution. We further show that the differential entropy of the velocity fields quantifies this asymmetry, with the classical information-geometric quantity $\mathbb{E}[\log |g'(X)|]$ emerging naturally as the entropy difference between cause and effect velocity fields. We introduce the R-score, a scale-normalized entropy criterion that identifies the root cause for any nonlinear mechanism via a Jensen gap argument. Empirically, the method achieves 99.3% accuracy on a synthetic bivariate causal discovery benchmark across 25 mechanism-distribution combinations and outperforms existing graph-based and graph-free baselines on the Sock-Shop microservice benchmark at AC@3 (82.4%) and AC@5 (90.8%), without requiring knowledge of the service dependency graph.

1 Introduction

The problem of root cause analysis (RCA) is to identify the source of an anomaly in a system comprising of multiple variables from observational data. What makes RCA challenging is that observed data could be explained by multiple causal structures, making it impossible to determine which variable is the source of the anomaly. As such, identifying the root cause directly from observational data remains ill-posed without imposing structural assumptions on the data generating process. To overcome the ill-posedness, various approaches have been proposed in the literature.

One possible approach to doing this is to restrict the class of functions connecting cause and effect variables and/or the class of noise distributions. The resulting asymmetry in the structural equation—where the model fits in one direction but not the reverse—enables identification of the causal direction from observational data alone. One of the earliest methods is the Linear Non-Gaussian Acyclic Model (LiNGAM) (Shimizu et al., 2006) which restricts the function class to linear mechanisms with non-Gaussian noise, recovering causal structure via independent component analysis. Subsequent methods such as additive noise models (Hoyer et al., 2008; Peters et al., 2014) and their post-nonlinear extensions (Zhang and Hyvärinen, 2009) relax linearity, establishing identifiability for nonlinear mechanisms by testing independence between regression residuals and the putative cause.

Other approaches exploit different sources of asymmetry. [Daniusis et al. \(2012\)](#); [Janzing et al. \(2012\)](#) leverage the postulate that cause and mechanism are independently chosen, using the log-Jacobian $\mathbb{E}[\log |g'(X)|]$ as a directional indicator without fitting models in both directions. [Rolland et al. \(2022\)](#) use the score function $\nabla \log p(x)$ as a causal probe, showing that the diagonal of its Jacobian identifies leaf nodes in a DAG, enabling topological ordering from a single observational distribution. [Peters et al. \(2016\)](#) exploit the stability of causal mechanisms across environments, using data from multiple conditions to identify causal variables. In the applied domain, RCA methods such as [Ikram et al., 2022](#)) and CIRCA ([Li et al., 2022](#)) typically assume or estimate a service dependency graph and rank candidate root causes via random walks or Bayesian scoring.

While these methods have significantly advanced causal inference, they face several challenges. LiNGAM’s identifiability relies on non-Gaussian noise, an assumption that may not hold in practice and whose verification requires first estimating the noise residuals under an assumed function class. ANMs require fitting models in both candidate directions and testing residual independence, a procedure whose reliability depends on the statistical power of the independence test, which degrades in high-dimensional systems with limited samples. Information-geometric methods such as IGGI overcome the need for assuming a specific function class, but rely on the independence postulate — that cause and mechanism are independently chosen — which can be violated when the input distribution and mechanism are coincidentally dependent ([Daniusis et al., 2012](#)).

In this work, we present a new source of asymmetry between cause and effect variables that arises when a system transitions from normal to faulty state, revealing the causal direction. More specifically, we show that the asymmetry arises in the velocity field of transporting samples from a “normal” distribution to the faulty one. In the present work, we employ optimal transport which selects the velocity fields that minimizes the kinetic energy of the transport. Due to the lack of a closed form for the velocity field, we use flow matching to learn them from observed data and subsequently prove that the resulting velocity field encodes the causal asymmetry: it factorizes in the causal direction and provably cannot factorize in the reverse for any nonlinear mechanism, yielding a new identifiability result without requiring a known causal graph, function fitting, or assumptions on the noise distribution. We further show that for stable systems, where perturbations attenuate along causal paths, this identifiability manifests as a simple scalar criterion: the cause variable exhibits higher velocity entropy than the effect. As a byproduct, the classical information-geometric quantity $\mathbb{E}[\log |g'(X)|]$ of ([Daniusis et al., 2012](#)) emerges naturally as the entropy difference between cause and effect velocity fields.

2 Methodology

We consider a joint collection of random variables $X_1, \dots, X_n$ governed by a directed acyclic graph (DAG) where causal relationships between the variables are represented by directed edges. The goal of RCA is to identify the variable at which a fault originates among these collection of variables $X_1, \dots, X_n$. To simplify the treatment, we consider the bivariate case and later show the generalization. We begin by considering random variables $X \rightarrow Y$ generated by a nonlinear additive noise model (ANM) given as,

$$Y = g(X) + \varepsilon_Y, \quad \varepsilon_Y \perp X \quad (1)$$

where $X \in \mathbb{R}$ is the cause, $Y \in \mathbb{R}$ is the effect, $g : \mathbb{R} \rightarrow \mathbb{R}$ is a nonlinear, differentiable mechanism, and ε_Y is additive noise independent of the cause and the mechanism. Causal inference methods built on this model class require additional identifiability assumptions on the mechanism and noise distribution (see [Peters et al. \(2014\)](#); [Montagna et al. \(2023\)](#)). Our approach presents new identifiability results requiring only nonlinearity of g , at the cost of observing the system under two operating conditions (i.e., normal and faulty) with mechanism invariance (Assumption 1).

Assumption 1. Under the ANM, we assume the existence of a normal and a faulty state such that the corresponding joint distributions of X, Y are specified as,

$$p_{(X,Y)}(x, y) = p_X(x)p_\varepsilon(y - g(x)) \quad (2)$$

$$q_{(X,Y)}(x, y) = q_X(x)p_\varepsilon(y - g(x)) \quad (3)$$

where p and q denote the distributions under normal and faulty states with $p_X \neq q_X$ and $p_Y \neq q_Y$ by virtue of fault propagation. Additionally, we consider the noise distribution p_ε and the mechanism g to be invariant across normal and faulty conditions.

Our approach to identifying the root cause is motivated by capturing the asymmetries of transforming cause and effect variables from normal to faulty states when an anomaly is injected. For the case of bivariate ANM, faults originating at the root cause (X) propagates to Y through the nonlinear mechanism. Since X is independent of the causal mechanism, transformation of the root cause from normal to faulty state is unconstrained. In contrast, the transformation of Y is constrained and mediated by the mechanism g , expanding and contracting the movement of Y during the transport from normal to faulty. From an information theoretic perspective, the constrained transformation causes the entropy of effect's transport to reduce as compared to that of the cause. We show that this asymmetry in the transformations reveals the causal direction and hence identify the root cause. To quantify the asymmetry in the transformations between cause and effect, and therefore perform RCA, we characterize how individual samples stretch and contract as they are transported from normal to faulty distributions. To this end, we leverage continuous normalizing flows (CNFs) which parametrizes the transformation as the solution of an ordinary differential equations (ODEs) governed by an unknown velocity field, modeled using a neural network. The velocity field captures the stretching and contraction of samples during the transport. More specifically, CNFs define a transport via an ODE

$$\frac{dx}{dt} = v_\theta(t, x), \quad \text{with } x(0) \sim p \text{ and } x(1) \sim q$$

where p and q are the source and target distributions and train the velocity field v_θ by maximum likelihood estimation. However, this requires simulating the ODE and its differentiation during training, making CNFs computationally expensive. More recently, [Lipman et al. \(2023\)](#) introduced flow matching that overcomes this limitation by prescribing a probability path p_t between the source and target distributions and training a neural network to regress directly onto the corresponding velocity field, yielding a simulation-free objective that is both fast and scalable. Since velocity fields define the transformation from normal to faulty states, we show that the resulting velocity fields encode the asymmetry in the transformation between the cause and effect variables.

2.1 Flow Matching

The premise of flow matching is to learn the velocity field that defines a time dependent mapping, referred hereafter a flow, $\psi : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ implementing $\psi : (t, x) \rightarrow \psi_t(x)$, to transform samples from the source distribution p to a target q . For some random variable $X_0 \sim p$, ψ_t generates a continuous time Markov process $(X_t)_{0 \leq t \leq 1}$

$$X_t = \psi_t(X_0), \quad t \in [0, 1]$$

with $\psi_0(X_0) \sim p$ and $\psi_1(X_0) \sim q$. Each sample traces a unique trajectory from p to q and is obtained as the solution of an ODE governed by a velocity field $v : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ implementing $v : (t, x) \mapsto v_t(x)$ as

$$\frac{d}{dt}\psi_t(z) = v_t(\psi_t(z)), \quad \psi_0(z) = z$$

Simulating this ODE starting with some initial condition, e.g., $X_0 = x_0 \sim p$ provides a powerful approach to generate samples from a target distribution q or any distribution p_t lying on the path from p to q . The resulting family of time-dependent probability distributions $(p_t)_{0 \leq t \leq 1}$ with $p_0 = p$ and $p_1 = q$ is referred to as a probability path and we say that the velocity field v_t generates the probability path p_t if $X_t = \psi_t(X_0) \sim p_t$ for all $t \in [0, 1]$. This establishes a working relationship between the velocity field, their flows, and the probability paths.

The process for learning the velocity field involves designing a probability path p_t interpolating the source and target distributions and then training the velocity field v_t^θ that generates p_t . Flow matching achieves this by minimizing a regression loss given as:

$$\mathcal{L}_{FM}(\theta) = E_{t, X_t \sim p_t(x)} \|v_t(X_t) - v_t^\theta(X_t)\|^2, t \sim U(0, 1). \quad (4)$$

An immediate obstacle in minimizing this loss is that neither the probability path nor the velocity field are unknown. In fact, multiple probability paths satisfies the boundary conditions, rendering the direct optimization of the loss function intractable. To overcome this, [Lipman et al. \(2023\)](#) showed that one can construct the marginal quantities $(p_t(x)$ and $v_t(x))$ by aggregating per sample probability paths and velocity fields each of which is obtained by conditioning on a target sample $X_1 = x_1$ denoted as $p_{t|1}(x|x_1)$ and $v_t(x|x_1)$. The marginal p_t is therefore obtained by integrating the conditional probability paths over the target distribution, i.e.,

$$p_t(x) = \int p_{t|1}(x|x_1)p_1(x_1)dx_1 \quad (5)$$

Boundary conditions $p_{0|1}(x|x_1) = p_{0,1}(x, x_1)/q(x_1)$ and $p_{1|1}(x|x_1) = \delta_{x_1}(x)$ consistent with the marginal probability path. Similarly, the velocity field v_t can be expressed as an aggregate of the conditional velocity fields $v_t(x|x_1)$, i.e.,:

$$v_t(x) = \int v_t(x|x_1)p_{t|1}(x_1|x)dx_1 \quad (6)$$

where $p_{1|t}(x_1|x)$ is given as follows using the Bayes' rule:

$$p_{1|t}(x_1|x) = \frac{p_{t|1}(x|x_1)q(x_1)}{p_t(x)} \quad (7)$$

and $v_t(x|x_1)$ is conditional velocity field.

Conditioning on a target sample allows constructing the marginal probability path and velocity fields. However, it still requires marginalizing over the entire training dataset, making it intractable. To overcome this problem, [\(Lipman et al., 2023\)](#) proposed Bregman divergence as the dissimilarity measure that allows learning v_t^θ using the conditional velocities $v_t(x|x_1)$, leading to a conditional flow matching loss as:

$$\mathcal{L}_{CFM} = E_{t, X_t, U \sim p_t|U} D(v_t(X_t|U), v_t^\theta(X_t))$$

and showed that the two losses are equivalent from the perspective of learning the parameters because their gradients coincide, i.e., $\nabla \mathcal{L}_{FM} = \nabla \mathcal{L}_{CFM}$ (see Theorem 4 in [\(Lipman et al., 2024\)](#)). With a simplified conditional flow matching loss, we still need to compute the conditional velocity fields. This is accomplished by defining a conditional flow $\psi_t(x|x_1)$ which defines both $p_t(x|x_1)$ and $v_{t|1}(x|x_1)$. [Lipman et al. \(2023\)](#) obtained the conditional flow as the minimizer of the dynamic optimal transport (OT) problem with quadratic cost, which is nothing but a linear interpolant given as, $X_{t|1} := \psi_t(X_0|x_1) = tx_1 + (1-t)X_0 \sim p_{t|1}$. The conditional velocity is then obtained by differentiating the linear interpolant, i.e.,

$$v_t(x|x_1) = \left. \frac{d}{dt} X_{t|1} \right|_x = \frac{x_1 - x}{1 - t} \quad (8)$$

Plugging in the formula for conditional velocity in the loss function, we obtain the following objective,

$$\mathcal{L}_{CFM}^{OT} = E_{t, X_0, X_1 \sim p_t | U} D((X_1 - X_0) - v_t^\theta(X_t)) \quad (9)$$

where $t \sim U(0, 1)$.

2.2 Velocity Field Asymmetry

With the flow matching machinery to learn the velocity fields of transporting samples from normal to faulty states, we now formalize the asymmetry that exists therein, lending to the identifiability of cause and effect. Considering the causal mechanism described by the ANM in Equation (1), we first show that the velocity fields of cause and effect variables are related to each other via the mechanism's Jacobian.

Lemma 1. *Let ϕ denote the OT map such that $\psi_t(\varepsilon) = (1 - t)\varepsilon + t\phi(\varepsilon)$. For two identically distributed random variables $\varepsilon_0 \sim p$ and $\varepsilon_1 \sim q$, the velocity field is 0.*

Proof. The Wasserstein distance between two probability distributions is the cost of the optimal transport problem, i.e., $\mathcal{W}(p, q) = \sqrt{\int \|\phi(\varepsilon_0) - \varepsilon_0\|^2 dp_{\varepsilon_0}}$. If $p = q$, $\mathcal{W}(p, q) = 0 \implies \int \|\phi(\varepsilon_0) - \varepsilon_0\|^2 dp_{\varepsilon_0} = 0$. This implies $\phi(\varepsilon_0) = \varepsilon_0$ almost surely. We can derive the velocity field of transporting p to q as,

$$v_t = \frac{d}{dt} \psi_t(\varepsilon_0) = \phi(\varepsilon_0) - \varepsilon_0 = 0 \quad (10)$$

□

Lemma 2. *Let causal mechanism be described by $Y = g(X) + \varepsilon_Y$ with $\varepsilon_Y \perp X$, where g is a differentiable mechanism. For some flow obtained via the conditional OT coupling, we have:*

$$v_Y = g'(X) v_X \quad (11)$$

Proof. For the pair (X, ε) , consider p_t^X and p_t^ε be probability paths generated by solving the optimal transport problem. We define random variable $X_t \sim p_t^X$ by taking a linear combination of $X_0 \sim p_X$ and $X_1 \sim q_X$ such that,

$$X_t = (1 - t)X_0 + tX_1 \implies \frac{dX_t}{dt} = x_1 - x_0 \quad (12)$$

Similarly, we define ε_t by taking the linear combination of noise in the normal and faulty states. But since they are identically distributed (Assumption 1), Lemma 1 shows that the velocity of transporting the noise variable is 0, hence

$$\frac{d\varepsilon_t}{dt} = 0 \quad (13)$$

Under mechanism invariance (Assumption 1), the structural equation $Y_t = g(X_t) + \varepsilon_t = g(X_t) + \varepsilon_0$ is maintained along the flow. Differentiating both sides with respect to time and using chain rule gives:

$$\frac{dY_t}{dt} = g'(X_t) \frac{dX_t}{dt} + \frac{d\varepsilon_Y}{dt} \quad (14)$$

Substituting $v_Y = \frac{dY}{dt}$, $v_X = \frac{dX}{dt}$, and $\dot{\varepsilon}_Y = \frac{d\varepsilon_Y}{dt} = 0$ yields:

$$v_Y = g'(X) v_X \quad \square$$

The relationship $v_Y = g'(X)v_X$ reveals how the mechanism mediates the transport of the effect variable. In particular when the mechanism is linear, i.e., $g'(X)$ is constant, effect's velocity is obtained by scaling the cause's velocity by a constant factor. The overall shape of the velocity field remains same, only the scale changes. In contrast, when the mechanism is nonlinear, $g'(X)$ varies across samples, causing the velocity field to stretch and compress. This non-uniform warping redistributes the probability mass in the velocity space. Regions with high $g'(X)$ stretches the velocity field, reducing the local density while low $g'(X)$ compresses the velocity field causing the local density to increase. In the following, we first show that the asymmetry in the velocity fields emerging due to non-linearity of the mechanism lends itself to the identification of cause and effect. We subsequently introduce differential entropy of the velocity fields to quantify this asymmetry and identify the root cause.

Theorem 1 (Velocity Identifiability). *Under Assumption 1, and non-zero velocity fields v_X and v_Y satisfying the velocity compatibility in Lemma 2 in the forward direction $X \rightarrow Y$ via a differentiable mechanism g as*

$$v_Y = g'(X_t)v_X$$

If the same velocity fields satisfies the velocity compatibility in the reverse direction $Y \rightarrow X$, i.e.,

$$v_X = h'(Y_t)v_Y$$

then g is affine. Conversely, if g is nonlinear, the velocity field cannot simultaneously satisfy the compatibility equations in both the directions.

Proof. Suppose the forward and reverse ANMs simultaneously hold. From the forward and reverse compatibility equations, we have

$$v_Y = g'(X_t)h'(Y_t)v_Y \quad (15)$$

Since the velocity field $v_Y \neq 0$, we get $g'(X_t)h'(Y_t) = 1$. Under non-trivial OT coupling, X_t traces the probability path generated by v_X such that $X_t|_{t=0} = x_0 \sim p$ and $X_t|_{t=1} = x_1 \sim q$. Similarly $Y_t = g(X_t) + \varepsilon$, along the trajectory. For any $X_t = x$, we have

$$g'(x)h'(g(x) + \varepsilon) = 1 \quad (16)$$

hold along every OT trajectory for every noise realization $\varepsilon \in \text{supp}(p_\varepsilon)$. Differentiating with respect to ε , we have $g'(x)h''(g(x) + \varepsilon) = 0$. Since $g(x)$ is continuously differentiable and $g'(x) \neq 0$. This implies $h''(g(x) + \varepsilon) = 0$. Since $g(x) + \varepsilon$ ranges in an open interval, we have $h''(y) = 0 \implies h(y) = ay + b$. Substituting the derivative of h in Equation (16),

$$g'(x)a = 1 \implies g(x) = \frac{x}{a} + c \quad (17)$$

In other words, g is affine if both forward and reverse ANMs are satisfied simultaneously. $\square$

This shows that the velocity field is not simultaneously compatible with both directions if the mechanism is non-linear, hence encoding information pertaining to the causal direction. While we have a machinery to compute the velocity fields, the mechanism is seldom known. To derive an analytical approach for RCA based on the asymmetry discussed in Theorem 1, we investigate the differential entropy of the velocity fields. Entropy is a natural choice to quantify the redistribution of the probability mass in the velocity space due to non-uniform stretching and compression by the mechanism's Jacobian. In regions where the Jacobian is large, local density decreases, increasing the entropy, while the entropy decreases in regions where $|g'(X)|$ is small due to local increase in the density. We show in the following that this effect is captured by the $E(\log |g'(X)|)$.

Lemma 3. *Under the ANM $Y = g(X) + \varepsilon$ and Assumptions 1, the differential entropy of the velocity fields satisfies:*

$$\mathcal{H}(v_Y) = \mathcal{H}(v_X) + E_{p_X}(\log |g'(X_t)|) \quad (18)$$

Proof. By Lemma 2 and under Assumptions 1, we have

$$v_Y = g'(X_t) \cdot v_X \quad (19)$$

where $g'(X_t)$ depends only on the mechanism. For a continuous random variable Z with probability density $p(z)$, the differential entropy is:

$$\mathcal{H}(Z) = - \int p(z) \log p(z) dz = -E(\log p(Z)) \quad (20)$$

Using change of variables, the density of v_Y is given as

$$p(v_Y) = p_{v_X} \left(\frac{v_Y}{g'(x)} \right) \left| \frac{1}{g'(x)} \right| \quad (21)$$

From the definition of differential entropy and change of variables, we have:

$$\begin{aligned} \mathcal{H}(v_Y) &= -E(\log p(v_Y)) \\ &= -E \left(\log p_{v_X} \left(\frac{v_Y}{g'(x)} \right) + \log \left| \frac{1}{g'(x)} \right| \right) \\ &= \mathcal{H}(v_X) + E(\log |g'(x)|) \end{aligned} \quad \square$$

Lemma shows that for a entropy contracting mechanism, i.e., $E(\log |g'(X)|) < 0$, the entropy of the effect's velocity field reduces when compared to the cause's velocity field. This behavior emerges from the underlying model assumptions. Given the cause X is independent of the mechanism g , any shift in the distribution of X originating from a fault is not constrained to g . On the contrary, the shift induced in the effect Y is mediated by the mechanism. As such, the velocity field for the cause, i.e., v_X is unconstrained and has higher entropy as compared to that of the effect. The following result states the identifiability result using the asymmetry in differential entropy.

Theorem 2. *Under the ANM, $Y = g(X) + \varepsilon_Y$ with $\varepsilon_Y \perp X$, Assumption 1, and $\dot{\varepsilon}_Y = 0$, if g is entropy-contracting under p_X , i.e., $E(\log |g'(X)|) < 0$, then:*

$$\mathcal{H}(v_X) > \mathcal{H}(v_Y) \quad (22)$$

and the root cause is identifiable as:

$$\hat{Z} = \arg \max_{Z \in \{X, Y\}} \mathcal{H}(v_Z) \quad (23)$$

Proof. From Lemma 2:

$$\mathcal{H}(v_Y) = \mathcal{H}(v_X) + \mathbb{E}[\log |g'(X)|]$$

Since g is entropy-contracting, $\mathbb{E}[\log |g'(X)|] < 0$, which directly gives $\mathcal{H}(v_Y) < \mathcal{H}(v_X)$. Therefore $\arg \max_Z \mathcal{H}(v_Z) = X$. $\square$

Example 1 Velocity asymmetry under entropy-contracting mechanism. To illustrate the velocity asymmetry under an entropy-contracting mechanism, we consider the ANM $Y = \tanh(X) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.05^2)$, where the mechanism $g(x) = \tanh(x)$ has Jacobian $g'(x) = \text{sech}^2(x) \in (0, 1]$ — strictly contracting everywhere. The normal state has $X \sim \mathcal{N}(1, 0.5^2)$, and a fault of magnitude $\delta = 1$ shifts the cause to $X \sim \mathcal{N}(2, 0.5^2)$ in the faulty state, while the noise realization remains

invariant ($\dot{\varepsilon} = 0$). We train a flow matching model to transport samples from the faulty state to the normal state and extract the velocity field at $t = 0$. Figure 1 shows the result: the cause velocity v_X is widely spread ($\mathcal{H}(v_X) = 0.91$), reflecting the unconstrained distributional shift at the root cause. In contrast, the effect velocity v_Y is compressed to a narrow band near zero ($\mathcal{H}(v_Y) = -0.93$), consistent with the mechanism’s Jacobian $|g'(X)| \approx 0.07$ attenuating the cause’s movement by a factor of approximately 15. The entropy gap $\mathcal{H}(v_X) - \mathcal{H}(v_Y) = 1.84$ correctly identifies X as the root cause.

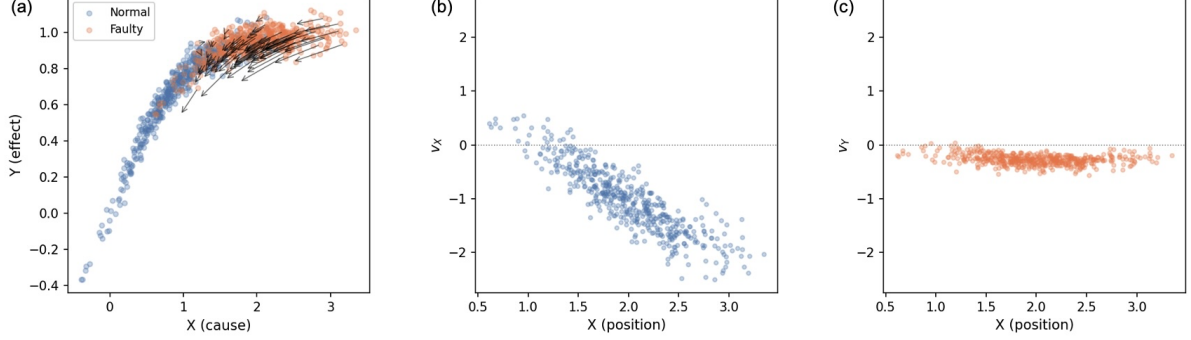


Figure 1: (a) Observed data under normal and faulty conditions along with the learned velocity fields transporting the samples from the faulty to normal state. Scatter plot of the (b) velocities of the cause and (c) velocities of the effect with respect to the cause.

One problem with the identifiability result in Theorem 2 is that it is only applicable when the mechanism is contracting. To generalize the velocity condition, we introduce a root cause score as follows.

Definition 1. For each variable Z in the causal DAG with velocity component v_Z , the root cause score is defined as,

$$\mathcal{R}(Z) = \mathcal{H}(v_Z) - \log \sigma_Z \quad (24)$$

where $\mathcal{H}(v_Z)$ is the differential entropy and $\sigma_Z = \text{std}(v_Z)$ is the standard deviation of the velocity values.

Assumption 2 (Jacobian-Velocity Independence). *The Jacobian of the mechanism is independent of the cause velocity:*

$$g'(X) \perp v_X \quad (25)$$

Lemma 4. *Under the ANM, $Y = g(X) + \varepsilon_Y$ with $\dot{\varepsilon}_Y = 0$, $g'(X) \perp v_X$:*

$$\sigma_Y^2 = \mathbb{E}[g'(X)^2] \sigma_X^2 + \text{Var}(g'(X)) \mu_{v_X}^2 \quad (26)$$

and therefore,

$$\log \sigma_Y = \log \sigma_X + \beta \quad (27)$$

where

$$\beta = \frac{1}{2} \log \left(\mathbb{E}[g'(X)^2] + \text{Var}(g'(X)) \frac{\mu_{v_X}^2}{\sigma_X^2} \right)$$

Proof. From Lemma 2 with $\dot{\varepsilon}_Y = 0$, we have $v_Y = g'(X) v_X$. The variance of v_Y is:

$$\sigma_Y^2 = \text{Var}(v_Y) = \mathbb{E}[g'(X)^2 v_X^2] - (\mathbb{E}[g'(X) v_X])^2 \quad (28)$$

By $g'(X) \perp v_X$ (Assumption 2), both expectations factor:

$$\begin{aligned} \mathbb{E}[g'(X)^2 v_X^2] &= \mathbb{E}[g'(X)^2] \mathbb{E}[v_X^2] \\ \mathbb{E}[g'(X) v_X] &= \mathbb{E}[g'(X)] \mathbb{E}[v_X] \end{aligned}$$

Substituting in the definition for σ_Y^2 , we have

$$\begin{aligned}\sigma_Y^2 &= \mathbb{E}[g'(X)^2] (\sigma_X^2 + E[v_X]^2) - (E[g'(X)v_X])^2 \\ &= \mathbb{E}[g'(X)^2] \sigma_X^2 + (E[g'(X)^2] - E[g'(X)]^2) \mu_{v_X}^2 \\ &= \mathbb{E}[g'(X)^2] \sigma_X^2 + \text{Var}(g'(X)) \mu_{v_X}^2\end{aligned}$$

Taking logarithms of both sides and dividing by two:

$$\log \sigma_Y = \log \sigma_X + \frac{1}{2} \log \left(\mathbb{E}[g'(X)^2] + \text{Var}(g'(X)) \frac{\mu_{v_X}^2}{\sigma_X^2} \right) \quad \square$$

Theorem 3. *Under the ANM, $Y = g(X) + \varepsilon_Y$ with Assumptions 1 – 2, we have,*

$$\mathcal{R}(X) - \mathcal{R}(Y) = \frac{1}{2} \log \left(\mathbb{E}[g'(X)^2] + \text{Var}(g'(X)) \frac{\mu_{v_X}^2}{\sigma_X^2} \right) - \mathbb{E}[\log |g'(X)|] \geq 0 \quad (29)$$

with equality if and only if g is affine. Therefore, $\arg \max_{Z \in \{X, Y\}} \mathcal{R}(Z) = X$ for any nonlinear g .

Proof. Let $\alpha = \mathbb{E}[\log |g'(X)|]$. From Lemma 3: $\mathcal{H}(v_Y) - \mathcal{H}(v_X) = \alpha$. From Lemma 4:

$$\log \sigma_Y - \log \sigma_X = \beta$$

Therefore:

$$\mathcal{R}(X) - \mathcal{R}(Y) = [\mathcal{H}(v_X) - \log \sigma_X] - [\mathcal{H}(v_Y) - \log \sigma_Y] \quad (30)$$

$$= (\log \sigma_Y - \log \sigma_X) - (\mathcal{H}(v_Y) - \mathcal{H}(v_X)) \quad (31)$$

$$= \beta - \alpha. \quad (32)$$

Since $\log$ is concave, Jensen's inequality gives:

$$\frac{1}{2} \log \mathbb{E}[g'(X)^2] \geq \mathbb{E}[\log |g'(X)|] = \alpha$$

Since, $\beta \geq \frac{1}{2} \log \mathbb{E}[g'(X)^2]$, we have $\beta \geq \alpha$ with equality iff $g'(X)$ is constant a.e., i.e., g is affine. $\square$

3 Implementation Details and Algorithm

We now describe the procedure for root cause identification using the asymmetry in velocity fields. The method consists of three stages: (i) pre-processing and normalization, (ii) learning the velocity field via conditional flow matching, and (iii) scoring each variable using the learned velocity components.

Given samples $\{x^{(i)}\}_{i=1}^{N_0} \sim p$ from the normal state and $\{x^{(i)}\}_{i=1}^{N_1} \sim q$ from the faulty state, where each $x^{(i)} \in \mathbb{R}^d$, we first standardize all variables using the statistics of the normal state:

$$\tilde{x}_j = \frac{x_j - \mu_j^{(p)}}{\sigma_j^{(p)}}, \quad j = 1, \dots, d \quad (33)$$

where $\mu_j^{(p)}$ and $\sigma_j^{(p)}$ are the mean and standard deviation of variable j under p . This normalization is applied to both normal and faulty samples, ensuring that the velocity field reflects genuine distributional shifts rather than differences in scale.

Next, to learn the velocity field of transporting samples from faulty to the normal state, we parameterize the velocity field using a neural network $v_t^\theta : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ and train it using

Algorithm 1 Flow Matching Root Cause Analysis

Require: Normal samples $\mathcal{D}_p = \{x_0^{(i)}\}_{i=1}^{N_0} \sim p$, faulty samples $\mathcal{D}_q = \{x_1^{(i)}\}_{i=1}^{N_1} \sim q$
Ensure: Root cause variable $\hat{Z}_{\text{RC}}$

- 1: **// Preprocessing**
- 2: Compute μ_j, σ_j from $\mathcal{D}_p$ for each variable j
- 3: Standardize $\mathcal{D}_p$ and $\mathcal{D}_q$ using (μ_j, σ_j)
- 4: **// Train flow matching model**
- 5: Initialize neural network v_t^θ
- 6: **for** epoch = 1, ..., E **do**
- 7: Sample $x_0 \sim \mathcal{D}_p$, $x_1 \sim \mathcal{D}_q$, $t \sim \mathcal{U}(0, 1)$
- 8: $x_t \leftarrow (1 - t)x_1 + tx_0$
- 9: $v^{\text{target}} \leftarrow x_0 - x_1$
- 10: Update θ to minimize $\|v_t^\theta(x_t) - v^{\text{target}}\|^2$
- 11: **end for**
- 12: **// Extract velocities and score**
- 13: **for** each $x_1^{(i)} \in \mathcal{D}_q$ **do**
- 14: $V^{(i)} \leftarrow v_0^\theta(x_1^{(i)})$ $\triangleright$ velocity at $t = 0$ (faulty state)
- 15: **end for**
- 16: **for** each variable $j = 1, \dots, d$ **do**
- 17: $\hat{\mathcal{H}}_j \leftarrow$ KDE entropy of $\{V_j^{(i)}\}_{i=1}^{N_1}$
- 18: $\sigma_j \leftarrow \text{std}(\{V_j^{(i)}\}_{i=1}^{N_1})$
- 19: **end for**
- 20: $\hat{Z}_{\text{RC}} \leftarrow \arg \max_j \hat{\mathcal{H}}_j$ $\triangleright$ H criterion (entropy-contracting)
- 21: *// Alternative:* $\hat{Z}_{\text{RC}} \leftarrow \arg \max_j (\hat{\mathcal{H}}_j - \log \sigma_j)$ $\triangleright$ R criterion (general)
- 22: **return** $\hat{Z}_{\text{RC}}$

the conditional flow matching objective. For the linear interpolation path from the faulty state ($t = 0$) to the normal state ($t = 1$):

$$x_t = (1 - t)x_1 + tx_0, \quad v_t^{\text{target}} = x_0 - x_1 \quad (34)$$

where $x_0 \sim p$ (normal) and $x_1 \sim q$ (faulty). The conditional flow matching loss is:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,1), x_0 \sim p, x_1 \sim q} \left\| v_t^\theta(x_t) - v_t^{\text{target}} \right\|^2 \quad (35)$$

After training the velocity field, we extract the velocity field at an arbitrary time period t along the probability path by evaluating the learned model on the faulty samples:

$$V = v_0^\theta(x_1), \quad x_1 \sim q \quad (36)$$

where $V \in \mathbb{R}^{N_1 \times d}$ contains the velocity values for each variable across all faulty samples. For each variable Z_j , $j = 1, \dots, d$, we compute the differential entropy of its velocity component $\mathcal{H}(v_{Z_j})$ using kernel density estimation:

$$\hat{\mathcal{H}}(v_{Z_j}) = -\frac{1}{N_1} \sum_{i=1}^{N_1} \log \hat{p}(v_{Z_j}^{(i)}) \quad (37)$$

where $\hat{p}$ is a Gaussian kernel density estimate fitted to $\{v_{Z_j}^{(i)}\}_{i=1}^{N_1}$. The root cause is identified as:

$$\hat{Z}_{\text{RC}} = \arg \max_{j \in \{1, \dots, d\}} \mathcal{H}(v_{Z_j}) \quad (38)$$

Table 1: Input distributions and mechanisms used in the Daniusis benchmark. All input distributions are supported on $[0, 1]$ via clipping.

Input distributions p_X		Mechanisms $g(x)$
A	$\mathcal{U}(0, 1)$	a $x^{1/3}$
B	$\mathcal{N}(0, \sigma^2)$	b $x^{1/2}$
C	$\mathcal{N}(0.5, \sigma^2)$	c x^2
D	$\mathcal{N}(1, \sigma^2)$	d x^3
E	$0.5\mathcal{N}(0.3, \sigma^2) + 0.5\mathcal{N}(0.7, \sigma^2)$	e $s_5(x)$

When the mechanism may be entropy-expanding, we instead use the scale-normalized score $\mathcal{R}(Z_j) = \mathcal{H}(v_{Z_j}) - \log \sigma_{Z_j}$, where $\sigma_{Z_j} = \text{std}(v_{Z_j})$.

The complete procedure is summarized in Algorithm 1. The dominant cost is training the flow matching model, which scales linearly with the number of variables d , the number of samples N , and the number of epochs E . The scoring step requires d kernel density estimations, each in $O(N_1^2)$ time. For systems with large d , the neural network architecture can be kept compact since the velocity field operates on the joint space and shares parameters across variables. The method requires no graph estimation, no bidirectional model fitting, and no independence testing — a single forward pass through the trained model yields the velocity components for all variables simultaneously.

4 Experiments

4.1 Simulated Benchmark: Bivariate Causal Discovery

We evaluate the proposed method on a synthetic benchmark inspired by the bivariate causal discovery setup of [Daniusis et al. \(2012\)](#). The benchmark tests root cause identification across 25 combinations of input distributions and causal mechanisms, providing a systematic assessment of the method’s robustness to different distributional and functional forms.

We consider the bivariate ANM $Y = g(X) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \lambda^2 \sigma^2)$, where a fault of magnitude δ is injected at the cause X . Five input distributions are paired with five nonlinear mechanisms. For each of the 25 combinations, we generate $N = 1000$ paired samples in normal and faulty states with fault magnitude $\delta = 1$, noise scale $\lambda = 0.1$, and $\sigma = 2$. We train a flow matching model (3-layer MLP, 128 hidden units, ELU activations) for 1000 epochs and identify the root cause using the velocity entropy criterion $\hat{Z}_{\text{RC}} = \arg \max_Z \mathcal{H}(v_Z)$. Each combination is repeated 100 times with independent random seeds to assess reliability.

Table 2 reports the root cause identification accuracy across all 25 combinations. The method achieves an overall accuracy of 99.3% (2483/2500 correct), with 23 out of 25 combinations achieving 100% accuracy. The two exceptions are $\mathcal{U}(0, 1)$ with $s_5(x)$ (90%) and $\mathcal{N}(0, \sigma^2)$ with $x^{1/2}$ (93%), both involving mechanisms where the Jacobian varies substantially across the input domain. The near-perfect accuracy across diverse input distributions and mechanisms — including both entropy-contracting ($x^{1/3}$, $x^{1/2}$, s_5) and entropy-expanding (x^2 , x^3 under certain distributions) mechanisms — demonstrates the robustness of the velocity-based approach. The method correctly identifies the root cause regardless of whether the mechanism contracts or expands the velocity field, consistent with the identifiability guarantees of Theorem 1. The slight accuracy reduction for $\mathcal{U}(0, 1)$ with $s_5(x)$ and $\mathcal{N}(0, \sigma^2)$ with $x^{1/2}$ can be attributed to the stochastic nature of the s_5 mechanism (randomly generated mixture of CDFs) and the near-singular behavior of $x^{1/2}$ near $x = 0$, where the Jacobian $g'(x) = \frac{1}{2\sqrt{x}} \rightarrow \infty$ introduces numerical instability in the velocity field estimation.

Table 2: Root cause identification accuracy (%) on the Danusis benchmark (100 repeats per combination). Overall accuracy: 99.3%.

p_X	$x^{1/3}$	$x^{1/2}$	x^2	x^3	$s_5(x)$
$\mathcal{U}(0, 1)$	100	100	100	100	90
$\mathcal{N}(0, \sigma^2)$	100	93	100	100	100
$\mathcal{N}(0.5, \sigma^2)$	100	100	100	100	100
$\mathcal{N}(1, \sigma^2)$	100	100	100	100	100
$\text{GM}([0.3, 0.7])$	100	100	100	100	100

4.2 Sock-Shop Microservice Benchmark

We evaluate the proposed method on the Sock-Shop microservice application from the RCD benchmark (Ikram et al., 2022), a widely used testbed for root cause analysis in cloud systems. The benchmark consists of fault injection experiments where a CPU hog is introduced at one of five services (carts, catalogue, front-end, orders, user), with five independent repetitions per service. Each incident provides time-series monitoring data for all services under normal and faulty conditions.

Table 3: Top- k recall of the proposed method (H criterion) on the Sock-Shop CPU hog benchmark. Results averaged over 5 repetitions $\times$ 10 random seeds.

Service	AC@1	AC@3	AC@5
carts	0.00	0.42	0.54
catalogue	0.46	0.84	1.00
front-end	0.30	0.94	1.00
orders	0.84	1.00	1.00
user	0.42	0.92	1.00
Overall	0.40	0.82	0.91

For each incident, we preprocess the monitoring data by applying a log-transformation ($\log(1 + x)$) to compress the dynamic range, injecting low-amplitude white noise (50 dB SNR) to regularize degenerate variables, and standardizing using normal-state statistics. We train a conditional flow matching model (3-layer MLP, 128 hidden units, tanh activations) for 10,000 steps per incident with batch size 64. The velocity field is extracted at $t = 0$ and the root cause is identified as $\hat{Z}_{\text{RC}} = \arg \max_Z \mathcal{H}(v_Z)$. Each of the 25 incident-scenario pairs (5 services $\times$ 5 repetitions) is repeated with 10 independent random seeds, yielding 250 total runs.

Table 3 reports the top- k recall for each service. The method achieves strong performance at AC@3 and AC@5, with overall recall of 82.4% and 90.8% respectively. Table 4 compares the proposed method against baselines reported in Ikram et al. (2022) on the four overlapping services (carts, catalogue, orders, user). Our method substantially outperforms all baselines at AC@3 and AC@5, achieving 79.5% and 88.5% respectively, compared to 60.0% and 60.0% for RCD, the strongest baseline. At AC@1, performance is comparable to RCD (43.0% vs 40.3%), with both methods significantly outperforming Ψ -PC and ε -Diagnoser.

The strong AC@3 and AC@5 performance demonstrates that the velocity field consistently ranks the true root cause among the top candidates, even when it is not the top-1 prediction. This is practically significant: in operational settings, engineers typically investigate a short list of candidates rather than relying on a single prediction. The weak performance on carts (AC@1 = 0%) warrants further investigation. One possible explanation is that the carts service has strong coupling with other services in the dependency graph, causing downstream variables

Table 4: Top- k recall comparison on Sock-Shop CPU hog for the four overlapping services (carts, catalogue, orders, user). Baseline results from [Ikram et al. \(2022\)](#). Best results in **bold**.

	Service	Ψ -PC	AutoMAP	ε -Diag.	RCD	Ours (H)
AC@1	carts	0.60	0.05	0.20	0.69	0.00
	catalogue	0.00	0.22	0.20	0.17	0.46
	orders	0.00	0.01	0.20	0.29	0.84
	user	0.00	0.29	0.20	0.46	0.42
AC@3	carts	0.60	0.23	0.40	0.86	0.42
	catalogue	0.00	0.22	0.20	0.22	0.84
	orders	0.00	0.36	0.20	0.66	1.00
	user	0.00	0.53	0.20	0.66	0.92
AC@5	carts	0.60	0.24	0.40	0.86	0.54
	catalogue	0.00	0.23	0.20	0.22	1.00
	orders	0.00	0.51	0.20	0.66	1.00
	user	0.00	0.55	0.20	0.66	1.00

to exhibit similarly large velocity shifts. Notably, RCD also struggles with certain services (catalogue: AC@5 = 0.22), suggesting that some fault patterns are inherently harder to localize. A key advantage of our method over the baselines is that it requires no knowledge of the service dependency graph. Ψ -PC, ε -Diagnoser, and RCD all estimate or assume a causal graph as a prerequisite, whereas our method operates directly on the joint monitoring data. Despite this, it achieves the highest overall recall at AC@3 and AC@5 among all methods.

5 Discussion and Conclusions

We introduced a new approach to root cause analysis based on the velocity field of optimal transport. By training a flow matching model to transport samples from a normal operating state to a faulty state, we showed that the learned velocity field encodes a fundamental asymmetry between cause and effect variables: the root cause transforms freely, while downstream variables are constrained by the causal mechanism. This asymmetry, visible as uniform transport paths for the cause and non-uniformly stretched or compressed paths for the effect, provides a new lens for causal inference that operates directly on distributional shifts without requiring a known causal graph, bidirectional model fitting, or independence testing.

The theoretical foundation rests on the observation that the optimal transport coupling in (X, ε) space naturally decomposes into transport of the cause and stationarity of the noise. This decomposition yields a clean velocity relationship $v_Y = g'(X_t)v_X$ (Lemma 2), connecting the velocity field directly to the causal mechanism’s Jacobian. We proved that this factorization can hold in only one direction for any nonlinear mechanism (Theorem 1), exploiting the fact that multiple OT trajectories carry different noise realizations, probing the reverse mechanism at different arguments and forcing it to be affine. This identifiability result is distinct from classical ANM identifiability ([Hoyer et al., 2008](#); [Peters et al., 2014](#)), which requires fitting models in both directions and testing residual independence. Our result instead derives from the structural properties of the velocity field under optimal transport, requiring only mechanism invariance and nonlinearity.

A noteworthy connection emerged between the velocity field entropy and the classical information-geometric quantity $\mathbb{E}[\log |g'(X)|]$ of [Daniusis et al. \(2012\)](#). IGCI extracts this quantity from a single observational distribution under the independence postulate, whereas our method extracts it from the distributional shift via the velocity field. The two approaches arrive

at the same asymmetry measure through different routes: IGCI through the independence of cause and mechanism, our method through the invariance of the mechanism across operating conditions. Building on this connection, we introduced the R-score $\mathcal{R}(Z) = \mathcal{H}(v_Z) - \log \sigma_Z$, which generalizes the entropy criterion to mechanisms that are not necessarily entropy-contracting, identifying the root cause for any nonlinear g via a Jensen gap argument.

Our method occupies a distinctive position in the RCA landscape. Graph-based methods such as RCD (Ikram et al., 2022) and CIRCA (Li et al., 2022) require estimating or assuming a service dependency graph, which is often infeasible in large-scale systems. Anomaly-scoring methods rank variables by their deviation from normal behavior but do not account for causal structure. Precision-matrix approaches (Schkoda and Janzing, 2025; Li et al., 2026) exploit covariance structure and work with single anomalous samples, but are restricted to linear structural equation models. Our method requires no graph knowledge, accounts for causal structure through the velocity field, and handles nonlinear mechanisms. Score-based causal discovery methods (Rolland et al., 2022; Montagna et al., 2023) also use learned vector fields to infer causal structure, but operate on a single observational distribution and recover topological ordering via the score’s Jacobian. In contrast, our method operates on the difference between normal and faulty distributions and uses the velocity field rather than the score function.

Empirically, the method achieves 99.3% accuracy on the Daniusis bivariate causal discovery benchmark across 25 combinations of input distributions and nonlinear mechanisms, spanning both entropy-contracting and entropy-expanding mechanisms. On the Sock-Shop microservice benchmark, the method achieves overall recall of 82.4% at AC@3 and 90.8% at AC@5, substantially outperforming all baselines including RCD (60.0% at both AC@3 and AC@5), without any knowledge of the service dependency graph. The strong performance at AC@3 and AC@5 is practically significant, as engineers typically investigate a short list of candidates rather than relying on a single prediction.

The present work has limitations that suggest directions for future research. The theoretical results are established for the bivariate case under the additive noise model. While the empirical results on the multivariate Sock-Shop benchmark are encouraging, extending the identifiability results to general DAGs with $d > 2$ variables, where the velocity of each node depends on the velocities of its parents through the local Jacobian, is the most pressing open problem. The method also requires multiple samples from both normal and faulty states, in contrast to recent single-outlier approaches (Schkoda and Janzing, 2025; Li et al., 2026). Looking forward, the connection between velocity field entropy and IGCI suggests applications beyond RCA, including general causal discovery from heterogeneous data and online root cause detection in streaming systems where distributional shifts are detected and diagnosed in real time.

References

- Povilas Daniusis, Dominik Janzing, Joris Mooij, Jakob Zscheischler, Bastian Steudel, Kun Zhang, and Bernhard Schölkopf. Inferring deterministic causal relations. In *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence*, pages 143–150, 2012.
- Patrik O Hoyer, Dominik Janzing, Joris M Mooij, Jonas Peters, and Bernhard Schölkopf. Nonlinear causal discovery with additive noise models. In *Advances in Neural Information Processing Systems*, volume 21, 2008.
- Azam Ikram, Sarthak Chakraborty, Subrata Mitra, Shiv Saez, Saurabh Veeraraghavan, and Erik Hermans. Root cause analysis of failures in microservices through causal discovery. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Dominik Janzing, Joris Mooij, Kun Zhang, Jan Lemeire, Jakob Zscheischler, Povilas Daniusis,

- Bastian Steudel, and Bernhard Schölkopf. Information-geometric approach to inferring causal directions. *Artificial Intelligence*, 182–183:1–31, 2012.
- Jinzhou Li, Benjamin B Chu, Ines F Scheller, Julien Gagneur, and Marloes H Maathuis. Root cause discovery via permutations and cholesky decomposition. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 88(3):779–798, 2026.
- Mingjie Li et al. Causal inference-based root cause analysis for online service systems with intervention recognition. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022.
- Yaron Lipman, Ricky T Q Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- Yaron Lipman, Marton Havasi, Peter Holderrieth, Neta Shaul, Matt Le, Brian Karrer, Ricky TQ Chen, David Lopez-Paz, Heli Ben-Hamu, and Itai Gat. Flow matching guide and code. *arXiv preprint arXiv:2412.06264*, 2024.
- Francesco Montagna, Nicoletta Noceti, Lorenzo Rosasco, Kun Zhang, and Francesco Locatello. Causal discovery with score matching on additive models with arbitrary noise. In *Conference on Causal Learning and Reasoning*, pages 726–751. PMLR, 2023.
- Jonas Peters, Joris M Mooij, Dominik Janzing, and Bernhard Schölkopf. Causal discovery with continuous additive noise models. *Journal of Machine Learning Research*, 15(58):2009–2053, 2014.
- Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: identification and confidence intervals. *Journal of the Royal Statistical Society: Series B*, 78(5):947–1012, 2016.
- Paul Rolland, Volkan Cevher, Matthaeus Kleindessner, Chris Russell, Dominik Janzing, Bernhard Schölkopf, and Francesco Locatello. Score matching enables causal discovery of nonlinear additive noise models. In *International Conference on Machine Learning*, pages 18741–18753, 2022.
- Daniela Schkoda and Dominik Janzing. Root cause analysis of outliers in unknown cyclic graphs. *arXiv preprint arXiv:2510.06995*, 2025.
- Shohei Shimizu, Patrik O Hoyer, Aapo Hyvärinen, and Antti Kerminen. A linear non-Gaussian acyclic model for causal discovery. *Journal of Machine Learning Research*, 7:2003–2030, 2006.
- Kun Zhang and Aapo Hyvärinen. On the identifiability of the post-nonlinear causal model. In *Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence*, pages 647–655, 2009.